

Pehan Amarathunge

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EDUCATION

University of Oxford, St John's College

October 2026 - September 2030

DPhil in Applied Mathematics

- Supervised by Prof. Ian Griffiths and Prof. Mohit Dalwadi.
- Funded via a Clarendon Scholarship

University of Cambridge, Girton College

October 2025 - June 2026

Master of Advanced Study (MASt) in Applied Mathematics (Part III)

Distinction

- **Relevant Modules:** Perturbation Methods, Fluid Dynamics of Climate, Non-Newtonian Fluids, Biological Fluids, Fluid Dynamics of Solid Earth, Slow Viscous Flows.
- **Essay** - *The Scallop Theorem*, supervised by Prof. Eric Lauga (92%).

University College London

September 2022 - June 2025

Bachelor of Science in Mathematics and Mathematical Physics

First Class Honors

- **Weighted Average** - 92.34 %
- **Rank** - 2/341
- **Relevant Modules:** Real Fluids (90.76 %), Geophysical Fluids (99.80 %), Asymptotic Methods

RESEARCH EXPERIENCE

Viscoelastic fluids in Medium Amplitude Oscillatory Shear (MAOS) (UCL)

Summer 2025

- Undertook a 8 week summer project **supervised by Prof. Helen Wilson** and funded by a UKRI Departmental Summer Studentship
- Investigated the theoretical underpinnings of a model equation by Dr. Peter Laity for the stress response of certain viscoelastic fluids under MAOS experiments
- Reviewed a novel framework by Randy Ewoldt for characterizing non-linear rheological behavior under an imposed MAOS strain.
- Investigated various constitutive relations in an attempt to match the empirical observations.

Lightning Solvers in Sloshing Modes (UCL)

Summers 2024/2025

- Undertook a 10 week summer project **supervised by Prof. Ted Johnson** and funded by the London Mathematical Society Undergraduate Research Bursary scheme.
- Generalized the Lightning Laplace solver (developed by Nick Trefethen) to deal with eigenvalue boundary conditions by solving a generalised eigenvalue problem instead of a least squares approach.
- Utilized this method to determine the sloshing modes and frequencies of a two-dimensional impermeable tank of arbitrary shape.
- Computations were performed on Matlab by modifying existing code written by Trefethen.
- Obtained up to 10 digits of accuracy, when comparing numerical results with existing analytical solutions for specific geometries.

AWARDS AND SCHOLARSHIPS

Clarendon Scholarship <i>University of Oxford</i> Scholarship to fund a DPhil at the University of Oxford.	October 2026
M.T. Meyer Taught Masters Achievement Scholarship <i>University of Cambridge, Girton College</i> An award of £400 for achieving a Distinction on Part III of the Mathematical Tripos.	June 2026
Ellen Watson Memorial Scholarship <i>UCL</i> Awarded to the best final year student in Applied Mathematics	October 2025
Dean's List <i>UCL Faculty of Mathematical and Physical Sciences</i> Awarded to the top 5% of graduating students in the faculty	October 2025
UKRI Summer Vacation Studentship <i>UKRI - UCL</i> Awarded a bursary of £2480 to fund a summer research project.	July 2025
London Mathematical Society - Undergraduate Research Bursary <i>UCL</i> Selected as one of 2 undergraduates in the UCL mathematics department and one of 32 mathematics undergraduates in the UK to receive an award of £2200 to undertake a research internship in the summer.	July 2024
Stevenson Prize <i>UCL</i> Awarded to the best first year student in applied mathematics out of a cohort of over 300 students	September 2023

TALKS AND CONFERENCES

1.Part III Seminar Series Cambridge Title: Viscoelastic Instabilities: Instability without Inertia	December 2025
2. British Conference of Undergraduate Research (BCUR) Newcastle Title: Lightning Solvers in Sloshing Modes	April 2024
3. Tomorrow's Mathematicians Today (TMT) Online Title: D'Alembert's Paradox and the Drag Crisis	March 2024

PROGRAMMING SKILLS

Proficient: Python, Matlab, $\text{\LaTeX}$

Python Libraries: NumPy, SciPy, Matplotlib, Pandas, Sympy

PROJECTS

Free Surface Flows Generated by a Moving Pressure Distribution (UCL) - Collaborated with a team of 4 undergraduates to investigate wave patterns generated by a moving pressure distribution, supervised by Dr. Sergei Timoshin. - Incorporated both gravity and capillary effects and presented our findings in a departmental colloquium.	July 2024 - August 2024
Addressing the Hare Eats Lynx Paradox (University of Colombo) - Undertook a 8 week summer project supervised by Dr. Kushani de Silva to review and generalise existing predator-prey models for the populations of Canadian Lynx and Snowshoe Hare, by including effects such as: fear, group defense, protection zones, hunting cooperation and epidemics. - Numerically simulated and compared these models using python and MATLAB to other models from other research papers and communicated my findings in weekly research meetings.	June 2023 - August 2023
Investigating the Stability of the Three Body Problem (UCL) - Liaised with a team of 4 undergraduates to compare the sensitivity of a three-body system to linear perturbations in various initial conditions using Python. - Presented our findings in a departmental symposium, implementing the Manim python library for animations and visualisations.	May 2023 - June 2023

TEACHING EXPERIENCE

Transition Mentor (UCL, Department of Mathematics)

September 2024 - December 2024

- Responsible for weekly meetings to help my tutees progress with problem sheets, extend their learning from lectures and prepare them for university examinations and the transition to university life
- Nominated as a finalist for "Transition Mentor of the Year" for my efforts.